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IT4010 Research Project – 2026 Jan

Topic Assessment Form

Project ID: R26-SE-032

1. Topic (12 words max)

An AI-Driven Decision Support System for Tomato Cultivation in Sri Lanka

2. Research group the project belongs to

CEAI - Centre of Excellence in AI

3. Specialization of the project belongs to

Software engineering

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Tomato farming in Sri Lanka faces several practical problems that affect crop yield, quality, and farmer income. Tomato growth depends heavily on soil conditions and environmental factors such as nutrient levels, moisture, and temperature [1]. Although soil test reports and basic weather data are available in many areas, farmers often find it difficult to understand these reports and apply them correctly in day-to-day farming. As a result, fertilizers and water are sometimes used incorrectly, leading to lower yields and poor-quality harvests [2]. Most existing agricultural tools only display raw data or provide general advice, which is not useful for specific locations or changing conditions. In addition, many farmers do not have access to advanced sensors or hardware-based systems.

Tomato diseases such as bacterial spot, tomato leaf curl virus, and early blight are another major challenge. Farmers mainly rely on visual inspection to identify these diseases, which often delays proper treatment [3] [4]. [5] [6]While some image-based tools are available, they usually only identify the disease and do not help farmers understand disease severity or evaluate whether treatments are effective. This leads to excessive chemical use, higher production costs, and negative impacts on human health and the environment.
[7] [8]Unstable market prices also create serious difficulties for tomato farmers. Prices

change frequently due to seasonal production, weather conditions, transport issues, and market demand. Small-scale farmers often depend on middlemen or personal experience when deciding when to harvest and sell their produce. Existing systems only show past or current prices and do not help predict future trends, resulting in income loss and increased post-harvest waste.

The main users affected by these problems are small- and medium-scale tomato farmers and agricultural officers in Sri Lanka. The impacts of the problem are social, economic, technical, environmental, and ethical, including reduced farmer income, inefficient farming practices, environmental damage, and unfair market conditions. This research falls under the **AgriTech domain** within the **Center of Excellence for Artificial Intelligence (CoEAI)** and aligns with the agriculture and food production industry. The proposed solution supports **SDG 2 (Zero Hunger)**, **SDG 8 (Decent Work and Economic Growth)**, **SDG 12 (Responsible Consumption and Production)**, and **SDG 13 (Climate Action)** by promoting sustainable, data-driven farming practices.

References

[1] J. Cañadas, J. A. Sánchez-Molina, F. Rodríguez and I. M. del Águila, "Improving automatic climate control with decision support techniques to minimize disease effects in greenhouse tomatoes," *Information Processing in Agriculture,* vol. 4, no. 1, p. 50–63, 2017.

[2] C. M. Flores Cayuela, R. González Perea, E. Camacho Poyato and P. Montesinos, "An ICT-based decision support system for precision irrigation management in outdoor and greenhouse tomato crops," *Agricultural Water Management,* vol. 269, p. 107686, 2022.

[3] H. S. a. H. S. Nagamani, "Tomato leaf disease detection using deep learning techniques.," *International Journal of Advanced Computer Science and Applications,* vol. 13, no. 1, 2022.

[4] H. B. V. P. K. M. N. S. K. T. S. R. S. a. D. S. K. S. J. Basha, "A Comprehensive Review on Deep Learning and Machine Learning Approaches for Tomato Leaf Disease Identification.," in *2023 8th International Conference on Communication and Electronics Systems (ICCES)*, Coimbatore, India, 2023.

[5] S. P. Mohanty,, . D. P. Hughes, and M. Salathé, , "Using Deep Learning for Image-Based Plant Disease Detection," *Frontiers in plant science,* p. 215232., 2016.

[6] Chéné, Yann, D. Rousseau, L. Philippe, Jessica Bertheloot, P. M. É. B. Valérie Caffier and F. Chapeau-Blondeau, "On the use of depth camera for 3D phenotyping of entire plants," *Computers and Electronics in Agriculture,,* vol. 82, pp. 122-127, 2012.

[7] Okere, E. E., & Balyan, V., "A Deep Learning-Based Prediction and Forecasting of Tomato Prices:," *A Deep Learning-Based Prediction and Forecasting of Tomato Prices for the Cape Town Fresh Produce Market:,* vol. 7(2), p. 19..

[8] Li, Zhe-Min, L. Guo Cui, Shi-Wei Xu, Ling-yun Weng, "Weekly Retail Prices," *"Prediction model of weekly retail price for the based on chaotic neural network.",* vol. 12, pp. 2292-2299, (2013) .

6. Brief description of Existing Research and Systems (300 words max)

Several existing research studies and agricultural systems have been developed to support farmers in improving crop productivity and management. Many agricultural decision support

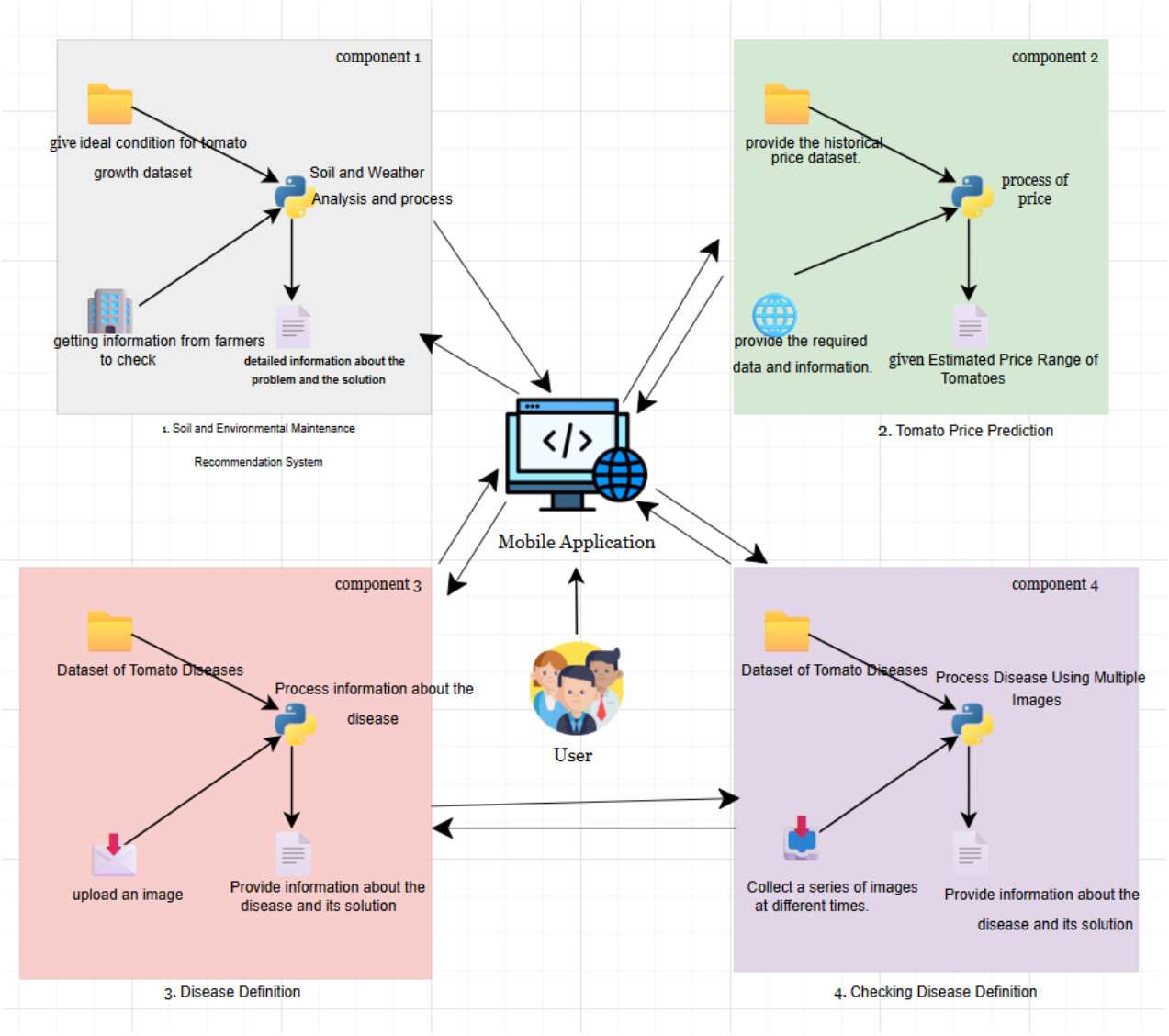
systems analyze soil nutrients, environmental conditions, or historical crop data to assist farming decisions. However, most of these systems mainly focus on data monitoring, visualization, or prediction models and do not provide clear, practical recommendations that farmers can easily apply. In addition, many solutions depend on IoT sensors and advanced hardware, which are not affordable or accessible for small-scale farmers.

In the area of tomato disease management, existing AgriTech applications and research systems commonly use image-based diagnosis supported by machine learning and deep learning techniques such as Convolutional Neural Networks (CNNs) and YOLO models. These systems are effective in detecting and classifying diseases such as bacterial spot, tomato leaf curl virus, and early blight. However, most studies focus on identifying a single disease under controlled conditions. They rarely assess disease severity, detect multiple diseases on the same leaf, or provide reliable treatment guidance and decision support for farmers.

Regarding market price analysis, some agricultural information networks and government platforms in Sri Lanka provide daily or weekly vegetable prices from major markets. These systems mainly act as price boards that display past or current prices and do not support future price prediction. Existing research has explored price forecasting using statistical models such as ARIMA and regression, as well as deep learning models like Long Short-Term Memory (LSTM). While these methods show good accuracy in controlled research settings, they are not implemented as practical, farmer-friendly tools tailored to the Sri Lankan context.

Overall, the main gaps identified include the lack of integrated, software-based decision support systems that combine soil and environmental analysis, multi-disease severity assessment with treatment evaluation, and localized price forecasting. Existing systems also lack post-treatment monitoring and actionable recommendations. This research aims to address these limitations by developing a simple, intelligent, and farmer-friendly solution suited to real-world agricultural use.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).



The proposed solution is an integrated, software-based decision support system designed to address the key gaps identified in existing research and systems. Unlike current solutions that focus on isolated problems such as raw data display, single-disease detection, or historical price viewing, this system combines soil and environmental analysis, tomato disease detection with severity assessment, treatment effectiveness monitoring, and market price forecasting into a single, farmer-friendly platform. The system does not rely on expensive IoT infrastructure and is designed to work with manually entered data and smartphone images, making it suitable for small- and medium-scale farmers in Sri Lanka.

How the Solution Addresses Identified Gaps

The system overcomes limitations of existing systems by providing:

- Actionable and explainable cultivation recommendations instead of raw data
- Multi-disease detection with disease-wise severity estimation
- Post-treatment monitoring to evaluate treatment effectiveness
- Localized tomato price forecasting to support better harvesting and selling decisions

System Perspective (Input – Process – Output)

Input

- Soil test report values (pH, Nitrogen, Phosphorus, Potassium)
- Manually entered environmental data (temperature, humidity, rainfall, wind speed)
- Multi-day datasets for trend analysis
- Tomato leaf images captured using smartphone cameras
- Historical tomato market price and supply data
- Optional regional market selection (e.g., Dambulla, Colombo)

Process

- Validation and preprocessing of user-entered data
- Comparison of soil and environmental values with ideal tomato growth thresholds
- Multi-day trend analysis using rule-based logic
- Image preprocessing and disease detection using deep learning models (CNN and YOLO)
- Disease severity estimation by identifying affected leaf regions
- Retrieval of verified treatment recommendations from a curated knowledge base
- Time-series analysis and price forecasting using machine learning models such as LSTM
- Post-treatment comparison of disease severity to assess treatment effectiveness

Output

- Tomato quality status (Good / Moderate / Poor)
- Quality and disease severity trends (Improving / Declining)
- Identified cultivation and disease-related issues
- Fertilizer, cultivation, and treatment recommendations
- Future tomato price predictions and price trend indicators
- Suggested optimal time periods for harvesting and selling
- Clear visual reports and notifications for farmers

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

   This project requires agricultural knowledge, technical skills, and reliable data to work successfully.

   Knowledge and Expertise

   Team members will learn about tomato farming, soil health, environmental needs, and diseases by reading guidelines, research papers, and talking to farmers and agricultural officers. They will also visit farms to see real tomato growth and disease conditions, which helps gain practical experience.

   Technical Skills

The project needs skills in rule-based decision systems, machine learning, deep learning, computer vision, and time-series analysis for price prediction. Team members will learn these skills through online tutorials, research papers, and small experiments on sample data. Practice projects and prototypes will help them gain confidence before building the full system.

Data Collection

Data will come from different sources:

- Soil data from laboratory test reports
- Environmental data (temperature, humidity, rainfall, wind) entered manually from weather records
- Tomato leaf images from public datasets, supplemented with photos collected from local farms
- Historical market prices and supply data from local markets

Optional IoT sensors can be used to collect real-time soil and environmental data. If sensors are not available, the system will still work using manual data. This ensures the project can be completed in all situations.

Ethical Considerations

The project does not involve humans or animals, so formal ethical approval is not needed. When collecting field data, the team will follow responsible practices, including getting permission from farm owners and protecting data privacy.

9. Objectives and Novelty

Main Objective To develop an intelligent decision support system that improves tomato cultivation quality using soil and environmental data.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Name: *Sivanesan Gabilan* Registration No: *IT22060426*	*To analyze soil test reports and environmental data and generate explainable recommendations to enhance tomato quality and productivity.*	*Tomato farmers often find it hard to grow healthy tomatoes with good quality. Problems happen because:* • *Soil types are different in each farm.* • *Weather changes (like temperature, rain, or humidity) affect growth.* • *Farmers may use the wrong amount or type of fertilizer.* *Most current methods only look at the data but do not give clear advice on what actions to take. This project helps farmers by analyzing soil tests and environmental data to give simple, easy-to-understand recommendations so they can grow better tomatoes.*	*The novelty of this project lies in its software-driven decision support approach that transforms manually entered soil test reports and environmental data into actionable, explainable recommendations. Unlike existing systems that focus only on data monitoring or require sensor-based infrastructure, the proposed solution evaluates quality trends over multiple days and provides area-specific guidance for improving tomato cultivation.*

		Inputs: 1. *Soil test reports (like pH, nutrients, and moisture levels)* 2. *Environmental data (like temperature, humidity, and rainfall)* 3. *Farmer information (like fertilizer used and watering schedule)* 4. *Data collected directly from farmers and agricultural officials* *Tasks / Process:* *Task 1: Understanding the Problem* • *Study how tomatoes are currently grown.* • *Find what is missing in current methods: they mostly monitor the data but do not give advice.* • *Decide what the system should do: analyze soil and environment, show trends over time, and give recommendations.* *Task 2: Collecting Data* • *Gather soil and weather data from different farms.* • *Collect information directly from farmers and agricultural officials.* • *Clean the data: remove mistakes, incorrect numbers, extra symbols, or irrelevant text.* • *Make data uniform: make all*	

		measurements and formats consistent (for example, soil pH always in numbers, temperatures in Celsius). • *Fix missing or wrong values so the dataset is complete and usable.* *Result: A ready dataset that can be analyzed by the system.* *Task 3: Preparing Data* • *Standardize all soil and environmental data so it can be compared easily.* • *Convert farmer-provided data into a common format.* • *Remove information that is not useful.* *Result: Clean and consistent data ready for analysis.* *Task 4: Choosing Important Factors* • *Decide which factors are most important for tomato growth, such as pH, nutrients (nitrogen, phosphorus, potassium), temperature, and humidity.* • *Combine some factors to show overall soil quality and environmental suitability.* *Result: A dataset that contains useful information to make recommendations.* *Task 5: Making Rules* • *Use simple agricultural rules to give advice:*	

		○ *If nitrogen is low → suggest nitrogen fertilizer.* ○ *If pH is too high or too low → suggest soil treatment.* ○ *Look at trends over several days → see if soil quality is improving or declining.* *Result: The system can give clear and specific actions to farmers.* *Task 6: Check Trends Over Days* • *Analyze soil and environmental data over multiple days.* • *Find patterns that affect tomato growth, such as consistent low moisture or nutrient deficiency.* *Result: Predictions and trends to help farmers plan better.* *Task 7: Give Recommendations* • *Suggest specific actions:* ○ *Change fertilizer type or amount* ○ *Adjust watering schedule* ○ *Treat soil if needed* • *Explain why each suggestion is given.* *Result: Farmers receive simple, easy-to-understand guidance.* *Task 8: Test and Validate* • *Check recommendations with*	

		farmers, agricultural officials, or past outcomes. • *Make sure the advice is practical and actually improves tomato growth and quality.* *Result: Recommendations are reliable and correct.* *Task 9: Make System Usable* • *Build a simple interface where farmers can enter their data.* • *Give real-time advice based on the entered data.* *Result: A working system that farmers can use to improve tomato cultivation.* *Outputs:* 1. *Trends of soil and environmental conditions over several days* ○ *The system shows how soil and weather conditions change over time.* ○ *For example, it can show if soil moisture is dropping, nutrients are decreasing, or temperature is too high for tomato growth.* ○ *This helps farmers see patterns and plan their actions, like when to water, fertilize, or treat the soil.* 2. *Clear advice on fertilizer,*	

		watering, and soil care ○ *The system gives specific, easy-to-follow suggestions for farmers.* ○ *Example recommendations:* ▪ *"Add nitrogen fertilizer because nitrogen level is low."* ▪ *"Water the plants twice a day because soil is too dry."* ▪ *"Use lime to adjust pH because soil is too acidic."* ○ *Each recommendation also explains why it is needed, so farmers understand the reason.*	

Name: *Vithusha Pathmanathan* Registration No: *IT22208262*	*To use intelligent decision support and image-based analysis to track the efficacy of tomato disease treatments.*	*Tomato growers use fungicides for early blight, leaf curl virus, and bacterial spot, but can't confirm the effectiveness of these treatments. A visual assessment overlooks minor improvements, resulting in wasted chemicals (LKR 50k/season), 20-30% reduction in yield, and soil contamination. Current tools identify diseases but do not assess severity reduction or offer alternatives when treatments are unsuccessful.* *Inputs:* ○ *Photos of tomato leaves (pre/post treatment)* ○ *Results of disease detection using Component 2 CNN* ○ *Applied treatment (name of fungicide, date)* ○ *Baseline images from Day 0 compared to re-scan images from Day 7/14* *Tasks/Process* *Task 1: Baseline recording of disease severity* ○ *Store these images as a baseline record in Firebase/JSON (or as a data*	*This project's closed-loop, software-driven decision support methodology, which assesses the efficacy of tomato disease treatments rather than ending at diagnosis, is what makes it novel. The suggested method objectively measures disease severity over time, incorporates environmental elements, and offers explainable alternative solutions without depending on IoT-based infrastructure, in contrast to current systems that only concentrate on disease detection or advisory services.*

		structure) which includes {plant identification, disease type, disease severity as a percentage, treatment applied, date recorded}. ○ *Utilize component 2, a convolutional neural network (CNN), as a U-Net structure for consistency in capturing disease severity at 55% during the baseline timeframe.* ○ *Include farmer identification and farm location for later review and comparison of results on multiple plants* ○ *Result: Data produced will be the standard baseline dataset against which treatment results are compared* *Task :2 Periodic Monitoring and Reassessment* ○ *Follow-up imaging was scheduled for 7 and 14 days after treatment.* ○ *Imaging was reprocessed utilizing an identical component 2 CNN uploaded into our U-Net format. The prognosis on post-processed images indicated*	

		25% severity at 7 days and 18% severity at 14 days. ○ Therefore, the cumulative recovery delta from baseline was 55%, the reduction in severity was 25%, and the difference was a 30% improvement. ○ *Overall results of time-series severity data reveal treatment responses and environmental impact.* *Task :3 Evaluating the Efficacy of Treatment* *Base vs Treatment Severity Assessment Outcome classification:* ○ *SUCCESS (greater than 20 percent): 55 percent to 25 percent* ○ *PARTIAL SUCCESS (5 to 20 Percent): 55 percent to 48 percent* ○ *FAILURE (less than 5 percent): 55 percent to 52 percent* ○ *Model Benchmark (U-Net): Accuracy of 95 percent compared to Human Visual: 60 percent.* ○ *Overall Goal: Verification using a data-driven method as opposed to a manual perspective.*	

		○ *Results: Objective classification of SUCCESS (30 percent reduction) based on quantifiable data to classify SUCCESS, PARTIAL FAILURE, and FAILURE.* *Task 4: An alternative treatment is recommended* ○ *if the crop does not respond to the treatment with copper (RAG Query), and the failure of treatment with copper will selfgenerate a prompt to look for another treatment via an alternative recommendation for treatment with "Mancozeb" at (2g/L once per week).* ○ *Alternative treatment recommendations are obtained by searching the database of alternative recommendations (the "DOA Knowledge Base") by querying on the basis of the crop condition or characteristics that are causing the problem.* ○ *There is an example of an alternative treatment recommendation to treat the leaf curl (an alternative to the*	

		alternative treatment with Mancozeb). For example, "remove all infected leaves and apply Imidacloprid at a labelled rate." ○ *A DOSAGE CALCULATOR - Calculate based on plant age (45 days), severity (52%) and number affected (15 total) for a total of 300 ml of solution to use for each of the 15 plants.* ○ *Only safe alternative treatment recommendations that have been reviewed and verified by the DOA (Department of Agriculture) for safe usage will be shown. The final results are: "Switch to Mancozeb, Alternative Treatment to Copper, (3% improvement), apply 2g/L to 15 plants."* ○ *This results in the avoidance of repeated failures of the same product when treated with another alternative product.* *Task :5 Data Visualization and Reporting* ○ *Farmer Friendly Presentation through Streamlit Dashboard*	

		○ *Using a line chart to show Severity trends from Day 0→7→14 (55%→25%→18%) and colorcoded zones* ○ *For status indicators, show GREEN as SUCCESS / YELLOW as PARTIAL / RED as FAILURE* ○ *Displaying multiple plant views as Plant #123, #145, and #167* ○ *Generating PDF reports with an example statement of "67% success rate out of 12 plants."* ○ *To convert complex AI plant analysis into farmer friendly insights that can be viewed at-aglance* ○ *Results are presented as a visual dashboard with downloadable reports showing Treatment Progress for All Monitored Plants*	

Name: *Rashad M. P. M* Registration No: IT22295842	*To find the optimal selling times for farmers using historical data on tomatoes and predict their future selling prices.*	*Tomato crop growers in Sri Lanka experience losses in income because of price variations influenced by seasonal changes, weather, and market demands. They depend on experience or middlemen for determining the time to sell, while the existing system does not enable future price prediction.* *This is remedied through the use of past prices and supply data to make an accurate forecast for short-term prices of tomatoes with guidelines for selling to the farmers.* *Inputs* • *Historical Price Data of Tomatoes in Sri Lanka (Daily/Weekly Market Price)* • *Data about tomato supplies (arrivals in markets, wherever available)* • *Time-related information: (Date, Month, Season, Holidays* • *Market location data (chief economic centers)* • *Data collected from government sources such as: the Department of Agriculture and the Economic Centers*	*The uniqueness of this component exists in the application of deep learning-based time series forecasting with localization of the predicted price of tomatoes in Sri Lanka through the use of LSTM. While common methodologies entail the use of average values or fixed price information, this solution predicts future market prices for a profitable time for the sale of the produce. The integration of the* *weekly prediction of the price of the produce into a computer-based decision support system enables the farmer to make proactive decisions.*

		Tasks / Process *Task 1: Understanding Pricing Problem* *Analysis:* • *Research the variation of tomato prices throughout the seasons and markets of Sri Lanka.* • *Determine critical farm issues being addressed:* • *Sudden price drops during peak harvest seasons* • *Lack of price forecasts* • *Dependence on middlemen* • *Define system objectives:* • *Forecast short-term tomato prices* • *Determine favorable periods of sale* • *Assist farmer sales choices* *Result: A clear definition of pricing and system goals.* *Task 2: Data Collection* *Get the historical price of tomatoes*	

		by: • *Dedicated Economic Centers* • *Agricultural government databases* *Gather information related to the supplies.* *• Merge the price and supply information into one dataset.* *Task 3: Cleaning and Preprocessing the Data* • *Remove records that are missing, duplicated, or incorrect.* • *Handle missing values through interpolation or averaging methods.* • *Use normalized price values to reduce the variation in scale.* • *Identify and handle seasonality patterns: monthly or seasonable trends.*	

		Task 4: Feature Extraction *List the important features like:* • *Previous day/week prices* • *Seasonal indicators (month, festive seasons)* • *Supply Variations* *Process data to ensure it can be converted to sequences for time series forecasts.* *Task5: Designing the LSTM Model* • *Implementing a Long Short-Term Memory (LSTM) neural network to learn the patterns of prices.* • *Train the model on the historical stock price and supply values.* • *Improve model parameters to yield better prediction.* *Task 6: Model Evaluation*	

		• *The model should be evaluated with accuracy metrics:* • *Root Mean Square Error (RMSE)* • *MAPE - Mean Absolute Percentage Error* • *Compare forecasted prices with actual prices.* *Task 7: Short-Term Price Forecasting* • *Short-term forecasts of tomato prices (for the coming 7 to 14 days).* • *Determine expected price rises and falls.* *Task 8: Decision Support and Recommendations* • *Analyze the predicted prices for the purpose of:* • *Best Time to Sell Best Time to Sell* • *Risk of price drops*	

		• *Formulate simple suggestions, for example:* *"Sell within the next 3 days – price expected to decrease."* *"Hold harvest for one week - Higher prices expected."* *Task 9: Integration of the System and its usability* • *To be incorporated into a decision support system.* • *Display prices and recommendations using:* • *Simple Plots* • *Simple messages* *Make sure that the system is available to farmers and agricultural officers.* *Outputs* • *Short-term tomato price prediction* • *Price trend graphics*	

		• *Selling opportunity indicators: high / medium / low* • *Sales recommendations to ensure maximum earnings to farmers*	

Name: *FARTHAS A L M* Registration No: IT22190734		*Collect and preprocess tomato leaf images for bacterial spot, tomato leaf curl virus, and early blight* *Train deep learning models (CNN / YOLO) to detect single and multiple diseases from leaf images* *Estimate disease-wise severity percentages using image analysis techniques* *Design a curated agricultural knowledge base containing verified treatment methods* *Integrate a Retrieval-Augmented Generation (RAG) mechanism to generate stage-appropriate treatment recommendations*	*The novelty of this component lies in its integrated approach that combines multi-disease detection, severity estimation, and knowledge- driven solution generation within a single system. Unlike existing tomato disease detection systems that focus only on single-disease classification, the proposed solution identifies multiple co-occurring diseases on the same leaf and quantifies their severity. Furthermore, the use of a RAG-based framework to retrieve verified treatment recommendations minimizes hallucinations and provides reliable, explainable solutions at the time of diagnosis.*

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Mr	Bimal	Gunapala	
Co-Supervisor	Mr.	Eishan	Weerasinghe	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				